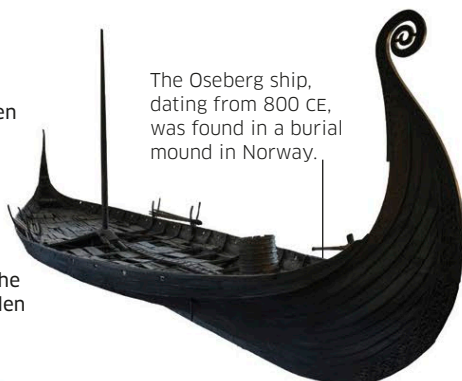


LASTING MATERIALS

Materials last for different lengths of time. Some materials decay in a matter of weeks, while some last for tens of thousands of years. The materials that survive for millennia provide a fascinating window into the way our ancestors used to live.

Viking long ship

Several Viking longships dating back more than a thousand years have been discovered intact in burial mounds. These ships were built of wood such as oak. Wood normally decays after a few hundred years, but the organisms that break it down need oxygen. There was no oxygen supply around the ships that lay buried, so the wood survived. Hulls of sunken wooden sailing ships survive underwater for the same reason.



The Oseberg ship, dating from 800 CE, was found in a burial mound in Norway.

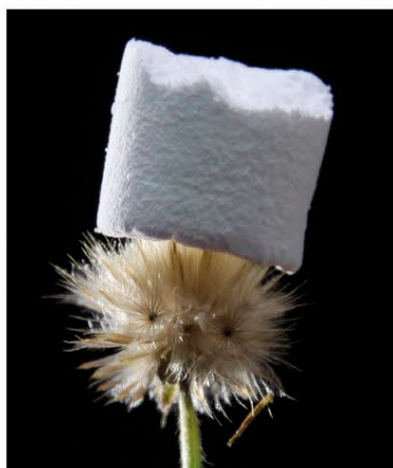


Roman amphitheater

Rome's Colosseum is made of several materials—a rock called travertine; another rock made of volcanic ash, called tuff; and concrete. It was built in 80 CE as an amphitheater. Since then it has been through wars and used as housing, factories, shops, and a fortress, but the basic materials have remained in place.

NEW MATERIALS

Material scientists—chemists, physicists, and engineers—research and deliver a steady stream of exciting new materials. Some materials resist damage, some heal themselves. There are plastics that conduct electricity, and wall coverings that reduce pollution. Environmental concerns are leading to materials designed to use fewer natural resources and to decompose without harmful waste.

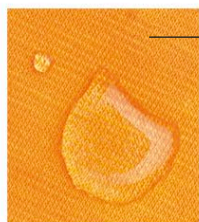


Aerogel

Aerogels are incredibly lightweight. Normal gels have a liquid and a solid component. In aerogels, the liquid is replaced by air—more than 99.8 percent of an aerogel is air. It protects from both heat and cold. Possible uses include insulation for buildings, space suits, and sponges for mopping up chemical spills.

Nanotechnology

Nanotechnology deals with materials that are between 1 and 100 nanometers wide or long. A nanometer is a millionth of a millimeter (making a housefly about 5 million nanometers long). This means new materials can be designed by moving and manipulating atoms.



This fabric is coated with water-repellent nanoparticles made of aluminium oxide.



The surface of a lotus leaf is naturally "nanostructured" to repel water.

REUSING AND RECYCLING

Reusing and recycling materials reduces the need to produce ever more of the materials we use a lot. This helps conserve raw materials, and cuts harmful carbon emissions. Materials production today considers the full life-cycle of a product—from reducing the raw materials and energy needed to produce it in the first place, to preventing materials from ending up in landfills and oceans.

The recycling sorting process

Different materials must be recycled in separate ways, and some things that get put in recycling bins cannot be recycled at all. The vast amount of materials we throw away gets processed in huge recycling centers.

